

# Existence of the TV-Wasserstein gradient flow

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September 2025



# TV Gradient Flow and TV-JKO Scheme

$J(\rho) = \int |\nabla \rho|$ , the total variation (TV) of  $\rho$ .

**TV-Gradient Flow:** 
$$\partial_t \rho_t + \operatorname{div} \left( \rho_t \nabla \operatorname{div} \left( \frac{\nabla \rho_t}{|\nabla \rho_t|} \right) \right) = 0$$

**TV-JKO Scheme:** 
$$\rho_{k+1}^\tau \in \operatorname{argmin} \left\{ \frac{1}{2\tau} W_2^2(\rho_k^\tau, \rho) + J(\rho) \right\}$$

where we viewed the term  $-\operatorname{div}(\frac{\nabla \rho}{|\nabla \rho|})$  as an element of  $\partial J(\rho)$ .

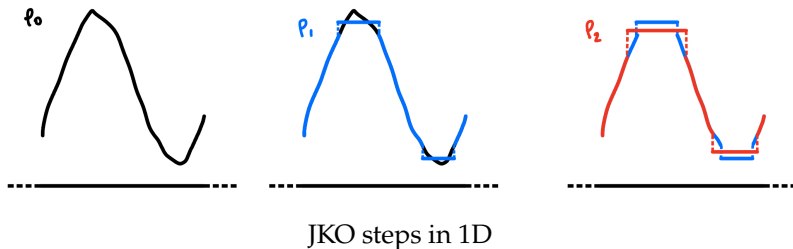
*Remark:*

$$\partial J(\rho) = \{ -\operatorname{div} z : z \in H_{\operatorname{div}}^1(\Omega), \|z\|_\infty \leq 1, J(\rho) = -\int_\Omega \rho \operatorname{div} z \}$$

Hereafter, we will write  $\operatorname{div} z$  to replace  $\operatorname{div}(\frac{\nabla \rho}{|\nabla \rho|})$ .

**What is this equation/JKO looks like?**

# Dimension 1



**Geometric view  $\implies$  :**

**Regularity  $\rho_k \implies$  :**

# General dimension: regularity of the level set

- regularity of  $\operatorname{div} z_1$ :

TV-JKO scheme:  $\rho_1 \in \operatorname{argmin} \left\{ \frac{1}{2\tau} W_2^2(\rho_0, \rho) + J(\rho) \right\}$

Euler-Lagrange equation:  $\frac{\varphi_1}{\tau} + \operatorname{div} z_1 = 0$ , a.e. in  $\operatorname{supp} \rho_1$

- $\operatorname{div} z \in \partial J(\rho) \implies \operatorname{div} z \approx \operatorname{div} \left( \frac{\nabla \rho}{|\nabla \rho|} \right)$  in the sense that  $\operatorname{div} z$  is the variational mean curvature for the level set of  $\rho$ .
- regularity of mean curvature  $\iff$  regularity of level set.

# Applied in Dithering reconstruction problem

Dithered image is a sum of Dirac masses



From left to right: Dithered image, TV-W and TV-L<sub>2</sub> results.  
Source: Chambolle, A., Duval, V., Machado, J.M. (2023)

# Return to the theoretical part

# Background

The JKO scheme has been studied by Carlier and Poon<sup>1</sup>. But the convergence of the JKO scheme was only presented in 1D.

*Why this restriction on the dimension ?*

Some estimates to guarantee compactness and pass the limit  $\tau \rightarrow 0$  require **lower bounds on the density**  $\rho_k^\tau$

**Upper bound**  $\int \rho^p, p \rightarrow \infty$

can be tackled via the usual flow interchange technique<sup>2</sup>.

**Lower bound:**  $\int \rho^p, p \rightarrow -\infty$

does not allow to use the flow interchange technique, since for  $p < 0$ , we do not have a displacement convex functional (except dimension  $n = 1$ ).

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<sup>1</sup>G. Carlier and C. Poon, On the total variation Wasserstein gradient flow and the TV-JKO scheme, ESAIM Control Optim. Calc. Var. 25 (2019), Paper No.42, 21, DOI 10.1051/cocv/2018042. MR4009553

<sup>2</sup>D. Matthes, R. J. McCann, and G. Savaré, A family of nonlinear fourth order equations of gradient flow type, Comm. Partial Differential Equations 34(2009), no. 10-12, 1352-1397, DOI 10.1080/03605300903296256. MR2581977



# What should we do ?

# What should we do ?

~~Give up and have a nice sleep with my cat~~



# If we consider the minimum principle in the PDE:

Let  $G(\mathbf{v}) = |\mathbf{v}|$ , we rewrite the equation as:

$$\partial_t \rho + \operatorname{div}(\rho \nabla \Delta_1 \rho) = 0$$

$$\text{where } \Delta_1 \rho := \operatorname{div}\left(\frac{\nabla \rho}{|\nabla \rho|}\right) = \nabla \cdot ((\nabla G)(\nabla \rho))$$

For any  $H$  convex, formally:

$$\begin{aligned} \frac{d}{dt} \int H(\rho) &= \int H'(\rho) \partial_t \rho \\ &= - \int H'(\rho) \nabla \cdot (\rho \nabla \Delta_1 \rho) \\ &= \int \rho H''(\rho) \nabla \rho \cdot \nabla \Delta_1 \rho \end{aligned}$$

Choose  $g : \nabla g(\rho) = \rho H''(\rho) \nabla \rho$

$$\begin{aligned} p = g(\rho) &\implies \Delta_1 \rho = \Delta_1 p \\ &= \int \nabla p \cdot \nabla \Delta_1 p \\ &= \int \sum_i p_i \left( \sum_j (G_j(\nabla p))_{ji} \right) \\ &= - \int D^2 p : D^2 G : D^2 p \leq 0. \end{aligned}$$

$\implies H(\rho)$  is decreasing along the flow, whenever  $H$  is convex.



# Minimal principle in the PDE and in the JKO steps

- **In JKO steps:** displacement convexity is mainly a technical requirement to obtain the minimum principle.
- **In the PDE:** displacement convexity is not required to obtain the minimal principle.
- **Difficulty:** We need a lower bound in the JKO level to obtain a solution of the PDE.

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## Our idea

- **At the JKO level,** add a small term to give an artificial lower bound at the JKO scheme level  
 $\implies$  get an approximated gradient flow.
- **At the gradient flow level,** pass the small term to 0.

# Main proof

- **At the JKO level**, add a small term to give an artificial lower bound at the JKO scheme level
- An approximated TV-JKO scheme:

$$\rho_0^\tau = \rho_0 \quad \rho_{k+1}^\tau \in \operatorname{argmin} \left\{ \frac{1}{2\tau} W_2^2(\rho_k^\tau, \rho) + \mathcal{F}_\epsilon(\rho) \right\}$$

$$\mathcal{F}_\epsilon(\rho) = J(\rho) + \int f_\epsilon(\rho), \quad f_\epsilon(s) = \begin{cases} \frac{\epsilon}{s-c} & \text{if } s > c \\ +\infty & \text{otherwise.} \end{cases}$$

- $\rho_\epsilon$  a weak solution of

$$\partial_t \rho + \operatorname{div} \left[ \rho \nabla \left( \operatorname{div} \left( \frac{\nabla \rho}{|\nabla \rho|} \right) - f'_\epsilon(\rho) \right) \right] = 0.$$

# Main proof

- **At the gradient flow level**, pass  $\epsilon \rightarrow 0$  to get the solution of the TV gradient flow.

$$\partial_t \rho + \operatorname{div} \left[ \rho \nabla \left( \operatorname{div} \left( \frac{\nabla \rho}{|\nabla \rho|} \right) - f'_\epsilon(\rho) \right) \right] = 0.$$

- A minimum principle for the approximated gradient flow.

Formal computation is rigorous thanks to:

- Strict convexity of  $f_\epsilon \implies \rho_\epsilon, f'_\epsilon(\rho_\epsilon) \in L_t^2 H_x^1$
- Domain  $\Omega = \mathbb{T}^d \implies$  no boundary terms.
- $\epsilon \rightarrow 0 \implies$  a weak solution of the TV gradient flow.

$$\partial_t \rho + \operatorname{div} \left[ \rho \nabla \operatorname{div} \left( \frac{\nabla \rho}{|\nabla \rho|} \right) \right] = 0.$$

## Theorem

- On the torus  $\mathbb{T}^d$
- the initial data  $\rho_0 \in \mathbb{P} \cap BV \cap L^\infty$  satisfies  $\rho_0 > c > 0$

the TV Wasserstein gradient flow admits a weak solution which remains bounded from below by  $c$ .



Thank you for your attention!